

Empirical Methods for Policy Evaluation

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Part 3: Difference-in-Differences and Job Search Models

- Two-way fixed effect regressions
 - ⇒ $DID \neq TWFE$
 - ⇒ Event Study regressions
 - ⇒ Heterogeneity-robust DID estimators
- Empirical job search models
 - ⇒ Eckstein-van den Berg (JoE, 2007)
- Informality, job search, and schooling investments
 - ⇒ Bobba-Flabbi-Levy (IER 2022)

DID $\neq$ TWFE

Groups and Time Periods

- We consider observations that can be divided into G groups and T periods
- For every $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$: = nb of obs in group g at period t
 - $\Rightarrow$ For simplicity, we assume hereafter balanced panel of groups:

$$\text{For all } (g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}, N_{g,t} > 0$$

- Panel/repeated cross-section data set where groups are, e.g.
 - $\Rightarrow$ Individuals, firms, counties, etc.
 - $\Rightarrow$ Cross-section dataset where cohort of birth plays the role of time
 - $\Rightarrow$ One may have $N_{g,t} = 1$, e.g. b/c group=individual or a firm

Potential Outcomes, SUTVA, and Covariates

- Let $(d_1, \dots, d_T)$ denote a treatment trajectory
- Corresponding **potential outcomes**: $Y_{g,t}(d_1, \dots, d_T)$
- Then **observed outcome**: $Y_{g,t} = Y_{g,t}(D_{g,1}, \dots, D_{g,T})$
- We maintain the usual **SUTVA**:

$$(Y_{g,1}(d_1, \dots, d_T), \dots, Y_{g,T}(d_1, \dots, d_T)) \perp (D_{g',t'})_{g' \neq g, t'=1, \dots, T}, \forall (g, t, d_1, \dots, d_T)$$

- For any scalar $X_{g,t}$, let $\mathbf{X}_g = (X_{g,1}, \dots, X_{g,T})$ and $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_G)$

The Simplest Setup, 2x2 Case

- Two groups $g \in \{s, n\}$ and two time periods $t \in \{1, 2\}$
- $D_{g,t} \in \{0, 1\}$, such that $D_{s,1} = D_{n,1} = 0$, $D_{s,2} = 1$, and $D_{n,2} = 0$
- $Y_{g,t}$ is the observed outcome in location g at period t
- $Y_{g,t}(0), Y_{g,t}(1)$ are potential outcomes without and with treatment

The Parallel (//) Trend Assumption

- In the absence of treatment, **same average outcome evolution** across groups

$$\mathbb{E}[Y_{s,2}(0) - Y_{s,1}(0)] = \mathbb{E}[Y_{n,2}(0) - Y_{n,1}(0)]$$

- Weaker than imposing that s and n have same untreated-outcome levels

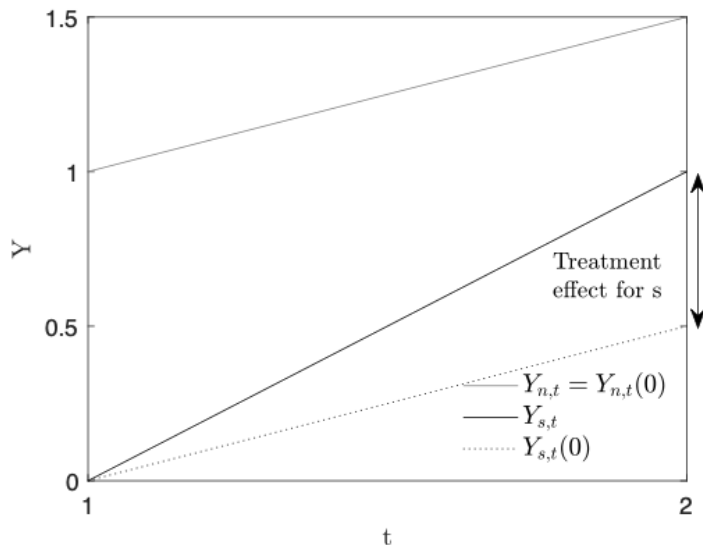
$$\mathbb{E}[Y_{s,t}(0)] = \mathbb{E}[Y_{n,t}(0)] \text{ for all } t$$

- Also weaker than imposing no variation in average untreated outcomes

$$\mathbb{E}[Y_{g,2}(0)] = \mathbb{E}[Y_{g,1}(0)] \text{ for all } g$$

- Appeal of // trends: has testable implications (**no pre-trends**)

Illustration of Parallel-trends



In the Simplest Set-up, TWFE = DID

- Under // trends, DID is unbiased for the ATE in location s at period 2

$$\begin{aligned}\mathbb{E}(DID) &= \mathbb{E}[Y_{s,2} - Y_{s,1} - (Y_{n,2} - Y_{n,1})] \\ &= \mathbb{E}[Y_{s,2}(1) - Y_{s,1}(0) - (Y_{n,2}(0) - Y_{n,1}(0))] \\ &= \mathbb{E}[Y_{s,2}(1) - Y_{s,2}(0)] + \mathbb{E}[Y_{s,2}(0) - Y_{s,1}(0)] - \mathbb{E}[Y_{n,2}(0) - Y_{n,1}(0)] \\ &= \mathbb{E}[Y_{s,2}(1) - Y_{s,2}(0)]\end{aligned}$$

- The before-after diff is combined with the treated-control diff
- This result extends beyond the 2x2 case as long as
 - $\Rightarrow$ Absorbing and binary treatment
 - $\Rightarrow$ No variation in treatment timing across groups (no staggered design)

The Pervasiveness of Two-way Fixed Effect Regressions

- Researchers often consider **two-way fixed effects** (TWFE) models of the kind:

$$Y_{g,t} = \alpha_g + \gamma_t + \beta_{fe} D_{g,t} + \epsilon_{g,t}.$$

- 26 out of the 100 most cited 2015-2019 AER papers estimate TWFE
- Also commonly used in other social sciences
- Other popular method: **event-study regressions**=dynamic version of TWFE

TWFE Parameters

- $\hat{\beta}_{fe}$ = OLS coeff. of $D_{g,t}$ in a reg. of $Y_{g,t}$ on group FEs, time FEs and $D_{g,t}$
- $\hat{\beta}_{fd}$ = OLS in a reg. of $Y_{g,t} - Y_{g,t-1}$ on time FEs and $D_{g,t} - D_{g,t-1}$
- $\hat{\beta}_{fe}^X$, when include covariates $X_{g,t}$ in the regression
- We first focus on β_{fe} , but we will extend the results to β_{fd} and β_{fe}^X

Additive Separability of TWFE

- A more general, yet **static**, case

$$D_{g,t} \in R^+ \text{ and for all } (g, t, d_1, \dots, d_T), Y_{g,t}(d_1, \dots, d_T) = Y_{g,t}(d_t)$$

- Parallel trends: for all $t \geq 2$, $E[Y_{g,t}(0) - Y_{g,t-1}(0)] = \gamma_t$
- It follows that: $E[Y_{g,t}(0) - Y_{g,1}(0)] = \gamma_t$, and let $\alpha_g = E[Y_{g,1}(0)]$
- Then, $E[Y_{g,t}(0)] = E[Y_{g,1}(0)] + E[Y_{g,t}(0) - Y_{g,1}(0)] = \alpha_g + \gamma_t$

Parameter of Interest

- Average treatment response

$$\Delta^{TR} = \frac{1}{\sum_{g,t} D_{g,t}} \sum_{g,t} (Y_{g,t}(D_{g,t}) - Y_{g,t}(0))$$

- Then, let $\delta^{TR} = E[\Delta^{TR}]$. With a **binary** D , $\delta^{TR} = \text{ATT}$
- Analogously, in (g, t) :

$$\Delta_{g,t} = \frac{1}{D_{g,t}} [Y_{g,t}(D_{g,t}) - Y_{g,t}(0)] \text{ if } D_{g,t} \neq 0$$

- Then:

$$\delta^{TR} = E \left[\sum_{(g,t): D_{g,t} > 0} W_{g,t} \Delta_{g,t} \right], \quad \text{with } W_{g,t} = \frac{D_{g,t}}{\sum_{(g,t): D_{g,t} > 0} D_{g,t}}$$

β_{fe} = weighted sum of ATEs under // trends

- Let $\epsilon_{fe,g,t}$ = resid. of the reg. of $D_{g,t}$ on a constant, group FEs, and time FEs
- It can be shown that:

$$\beta_{fe} = E \left[\sum_{(g,t): D_{g,t} > 0} W_{fe,g,t} \Delta_{g,t} \right]$$

$$\Rightarrow W_{fe,g,t} = \frac{D_{g,t} \epsilon_{fe,g,t}}{\sum_{(g,t): D_{g,t} > 0} D_{g,t} \epsilon_{fe,g,t}}$$

- In general, $\beta_{fe} \neq \delta^{TR}$ because $W_{fe,g,t} \neq W_{g,t}$
 - $\Rightarrow$ We may have $W_{fe,g,t} < 0$: if $\epsilon_{fe,g,t} < 0$ while $D_{g,t} > 0$
 - $\Rightarrow E[\hat{\beta}_{fe}]$ may be < 0 even if $Y_{g,t}(d) > Y_{g,t}(0)$ for all (g,t) and $d > 0$

Characterizing (g, t) cells weighted negatively by β_{fe}

- Let $D_{g,.}$ = average treat. rate of g and $D_{.,t}$ = average treat. rate at t
- Under // trends, $W_{fe,g,t}$ is decreasing with $D_{g,.}$ and $D_{.,t}$
- β_{fe} more likely to assign negative weight to:
 - $\Rightarrow$ Periods with many vs few treated groups
 - $\Rightarrow$ Groups treated for many vs few periods
- In staggered adoption designs ($D_{g,t} \geq D_{g,t-1}$), $W_{fe,g,t} < 0$ more likely
 - $\Rightarrow$ In the last periods and for groups adopting the treatment earlier
 - $\Rightarrow$ Remove negative weights by removing always treated groups

Forbidden Comparison 1: Switchers Vs. Always Treated

- *D* binary and design staggered: $\hat{\beta}_{fe}$ = weighted avg of two types of DID:
 - $\Rightarrow DID_1$: comparing s from untreated to treated to n untreated at both dates
 - $\Rightarrow DID_2$: comparing switching group s to group a treated at both dates.
- Negative weights in β_{fe} originate from the second type of DID

Forbidden Comparison 1: An Example

- Group e treated at $t = 2$, group ℓ treated at $t = 3$. Then:

$$\hat{\beta}_{fe} = \frac{1}{2} \times \underbrace{DID_{e-\ell}^{1-2}}_{DID_1} + \frac{1}{2} \times \underbrace{DID_{\ell-e}^{2-3}}_{DID_2}$$

- At periods 2 and 3, e 's outcome = treated potential outcome, so

$$Y_{e,3} - Y_{e,2} = Y_{e,3}(1) - Y_{e,2}(1) = Y_{e,3}(0) + \Delta_{e,3} - (Y_{e,2}(0) + \Delta_{e,2}).$$

- On the other hand, group ℓ only treated at period 3, so

$$Y_{\ell,3} - Y_{\ell,2} = Y_{\ell,3}(0) + \Delta_{\ell,3} - Y_{\ell,2}(0)$$

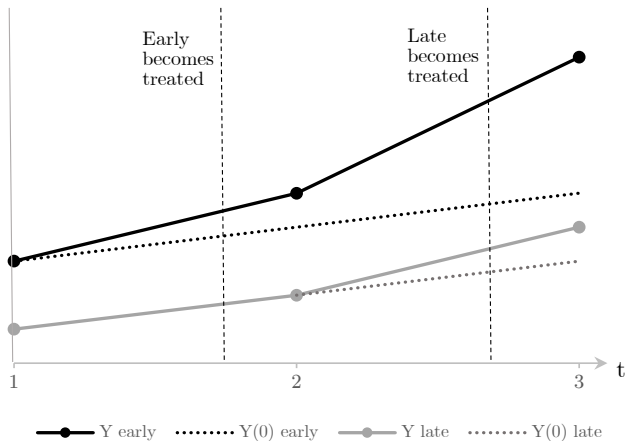
Forbidden Comparison 1: An Example (continued)

- $E[DID_{\ell-e}^{2-3}] = E[Y_{\ell,3} - Y_{\ell,2} - (Y_{e,3} - Y_{e,2})] = E[\Delta_{\ell,3} + \Delta_{e,2} - \Delta_{e,3}]$
- Note: if $\Delta_{e,2} = \Delta_{e,3}$, $E[DID_{\ell-e}^{2-3}] = E[\Delta_{\ell,3}]$
- More generally, if $\Delta_{g,t} = \Delta_{g,t'}$, $W_{fe,g,t} \geq 0$. But restrictive!
- Note:

$$Y_{g,t}(0) - Y_{g,t-1}(0) = Y_{g,t}(1) - Y_{g,t-1}(1) \iff \Delta_{g,t} = \Delta_{g,t-1}$$

- This assumption is actually equivalent to **time-invariant treat. effects**

Forbidden Comparison 1: Graphical Illustration



Forbidden Comparison 2: Switching more vs Switching less

- Suppose the treatment D is not binary
- $\hat{\beta}_{fe} = (\text{group } m \text{ whose } D \text{ increases more}) - (\text{group } \ell \text{ whose } D \text{ increases less})$
- In fact, with two groups m and ℓ and two periods,

$$\hat{\beta}_{fe} = \frac{Y_{m,2} - Y_{m,1} - (Y_{\ell,2} - Y_{\ell,1})}{D_{m,2} - D_{m,1} - (D_{\ell,2} - D_{\ell,1})}$$

- This “Wald-DID” estimator may not estimate convex combination of effects

Forbidden Comparison 2: An Example

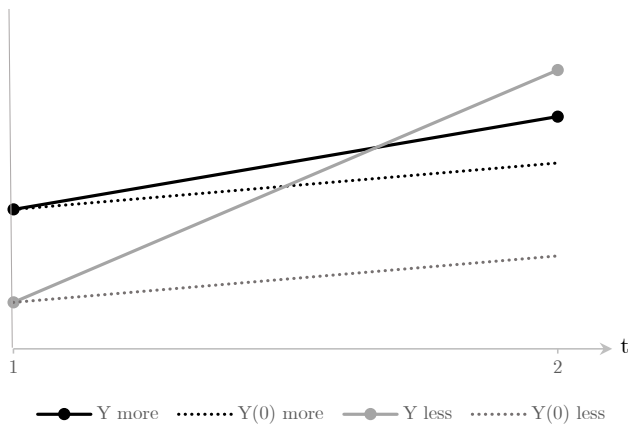
- Assume m goes from 0 to 2 units of treatment while ℓ goes from 0 to 1

$\Rightarrow$ Denominator of the Wald-DID is $2 - 0 - (1 - 0) = 1$

- Potential outcomes: $Y_{m,t}(d) = Y_{m,t}(0) + \delta_m d$ and $Y_{\ell,t}(d) = Y_{m,t}(0) + \delta_\ell d$
- Then:

$$\begin{aligned}
 E \left[\widehat{\beta}_{fe} \right] &= E \left[Y_{m,2} - Y_{m,1} - (Y_{\ell,2} - Y_{\ell,1}) \right] \\
 &= E \left[Y_{m,2}(0) + 2\delta_m - Y_{m,1}(0) - (Y_{\ell,2}(0) + \delta_\ell - Y_{\ell,1}(0)) \right] \\
 &= E \left[Y_{m,2}(0) - Y_{m,1}(0) \right] - E \left[Y_{\ell,2}(0) - Y_{\ell,1}(0) \right] + 2\delta_m - \delta_\ell \\
 &= 2\delta_m - \delta_\ell
 \end{aligned}$$

Forbidden Comparison 2: Graphical Illustration



Extensions

- This logic extends to β_{fd} , but with different weights $W_{fd,g,t}$

$\Rightarrow$ If $\beta_{fd} \neq \beta_{fe}$, we **reject homogeneous TE** under // trends

- With **covariates**, we modify the // trends by assuming that for some λ ,

$$\begin{aligned} & E[Y_{g,t}(0) - Y_{g,t-1}(0) - (X_{g,t} - X_{g,t-1})'\lambda | \mathbf{D}_g, \mathbf{X}_g] \\ &= E[Y_{g,t}(0) - Y_{g,t-1}(0) - (X_{g,t} - X_{g,t-1})'\lambda], \end{aligned}$$

- Let $\epsilon_{fe,g,t}^X$ = resid. of reg. of $D_{g,t}$ on a const., group FEs, time FEs and $X_{g,t}$
- Then, **same result as above** but with $\epsilon_{fe,g,t}^X$ instead of $\epsilon_{fe,g,t}$ in $W_{fe,g,t}$.

Software Implementations

- `bacondecomp` computes DIDs and their corresponding weights entering in $\hat{\beta}_{fe}$
- `twowayfeweights` computes the weights $W_{fe,g,t}$ and $W_{fd,g,t}$, with/o X
 - ⇒ Worst-case scenario of std dev on $\Delta_{g,t}$ (weights are max corr. with TE)
 - ⇒ Correlation between weights and proxies of $\Delta_{g,t}$

Example: Effect of Newspapers on Electoral Turnout

- Gentzkow et al. (AER, 2011) use US county data on presidential elections
- Reg **change in turnout** in g on **change in # newspapers** + state-year FE
- One could also estimate the FE regression

	$\hat{\beta}$	% of < 0	Sum of < 0
Regression	(s.e.)	weights	weights
$\hat{\beta}_{fe}$	-0.0011 (0.0011)	40.1%	-0.53
$\hat{\beta}_{fd}$	0.0026 (0.0009)	45.7%	-1.43

$\Rightarrow$ Under // trends, we **reject the null** hypothesis that $\Delta_{g,t} = \Delta \forall (g, t)$

Event Study Regressions

Main Assumptions

- Assuming past treatments do not affect current outcomes is restrictive
- We now generalize the previous set-up as follows ($\mathbf{0}_t$ = vector of t zeros)
- Univariate, **dynamic framework w/o anticipation**, $D_{g,t} \in R^+$:

$$Y_{g,t}(d_1, \dots, d_T) = Y_{g,t}(d_1, \dots, d_t)$$

- // trends, v2**

$$\Rightarrow \forall g \text{ and } t \geq 2, E[Y_{g,t}(\mathbf{0}_t) - Y_{g,t-1}(\mathbf{0}_{t-1}) | \mathbf{D}_g] = E[Y_{g,t}(\mathbf{0}_t) - Y_{g,t-1}(\mathbf{0}_{t-1})]$$

$$\Rightarrow \text{For all } t \geq 2, E[Y_{g,t}(\mathbf{0}_t) - Y_{g,t-1}(\mathbf{0}_{t-1})] = \gamma_t$$

Staggered Design: Notation and Parameters

- **Staggered treatment:** $D_{g,t} = 1\{t \geq F_g\}$, F_g = period when g becomes treated
- Exclude always treated here and let $F_g = T + 1$ if never treated
- Let $\mathbf{1}_t$ be a vector of t ones and define:

$$\Delta_g(\ell) = Y_{g,F_g+\ell}(\mathbf{0}_{F_g-1}, \mathbf{1}_{\ell+1}) - Y_{g,F_g+\ell}(\mathbf{0}_{F_g+\ell}),$$

$$\Delta(\ell) = \frac{1}{N_{\leq}(T-\ell)} \sum_{g:F_g+\ell \leq T} \Delta_g(\ell)$$

where $N_{\leq}(t) = \text{card}\{g : F_g \leq t\}$

$\Rightarrow \Delta_g(\ell)$: effect for g of having received the treatment for $\ell + 1$ periods

$\Rightarrow \Delta(\ell)$: effect of having received the treatment for $\ell + 1$ periods

Staggered design: ES Regressions

- Regression of $Y_{g,t}$ on group and time FEs, and $(1\{F_g = t - \ell\})_{\ell=-K,\dots,L, \ell \neq -1}$
 - When $t - \ell > T$, we set $1\{F_g = t - \ell\} = 0$
- ⇒ For $\ell \geq 0$, β_ℓ is supposed to estimate $\Delta(\ell)$
- ⇒ For $\ell \leq -2$, $\beta_\ell =$ placebo used to check // trends

An Aside: “Basic” Problems with ES Regressions

- Let us consider the **fully-dynamic specification** where

$$K = \overline{K} := \max_{g: F_g \leq T} F_g - 1, \quad L = \overline{L} := T - \min_g F_g$$

- If all groups are eventually treated, then:
- $(\beta_\ell)_{\ell=-\overline{K}, \dots, \overline{L}, \ell \neq -1}$ **not sep. identified** from $(\beta_\ell + \kappa(\ell + 1))_{\ell=-\overline{K}, \dots, \overline{L}, \ell \neq -1}$

$$\begin{aligned} \Rightarrow \text{Proof: } & \sum_{\substack{\ell=-\overline{K} \\ \ell \neq -1}}^{\overline{L}} 1\{F_g = t - \ell\}(\ell + 1) = \sum_{\ell=-\overline{K}}^{\overline{L}} 1\{F_g = t - \ell\}(\ell + 1) \\ &= (t + 1 - F_g) \sum_{\ell=-\overline{K}}^{\overline{L}} 1\{F_g = t - \ell\} \\ &= \underbrace{t + 1}_{\text{enter in time FE}} \underbrace{- F_g}_{\text{enter in group FE}} \end{aligned}$$

An Aside: “Basic” Problems with ES Regressions

⇒ The fully-dynamic specification requires **never treated groups**

- Otw some placebo coeffs should be removed, or dyn. effects be restricted
- In the latter case, a common practice is to choose $L < \bar{L}$
- But even if TE are homogenous, this makes sense only if

$$Y_{g,F_g+\ell}(\mathbf{0}_{F_g-1}, \mathbf{1}_{\ell+1}) = Y_{g,F_g+\ell}(\mathbf{0}_{F_g}) \quad \text{for } \ell > L.$$

- More natural to assume that TE stabilize:

$$\Delta_g(\ell) = \Delta_g(L)(= \Delta(L)), \quad \text{for } \ell > L$$

Decomposition Result for ES Regressions

- $\epsilon_{g,t}$: res. of $1\{F_g = t - \ell\}$ on group/time FEs and $(1\{F_g = t - \ell'\})_{\ell'=-K,\dots,\bar{L}}$
- Define $W_{g,\ell,\ell'} = \frac{\epsilon_{g,F_g+\ell'}}{\sum_g \epsilon_{g,F_g+\ell}}, \forall(\ell, \ell')$
- Suppose that $D_{g,t}$ is binary and design is staggered. Then, for $\ell = 0, \dots, \bar{L}$,

$$E[\hat{\beta}_\ell] = E \left[\underbrace{\sum_g W_{g,\ell,\ell} \Delta_g(\ell)}_{\text{sum of effect of } \ell + 1 \text{ treat. periods}} + \underbrace{\sum_{\ell' \neq \ell} \sum_g W_{g,\ell,\ell'} \Delta_g(\ell')}_{\text{sum of effects of } \ell' + 1 \text{ treat periods}} \right]$$

- $\sum_g W_{g,\ell,\ell} = 1$ and $\sum_g W_{g,\ell,\ell'} = 0 \forall \ell, \ell' \neq \ell$

ES Even Less Robust to Heterog. TE than Static TWFE

- As in the static case, $W_{g,\ell,\ell}$ may be < 0

$\Rightarrow \hat{\beta}_\ell$ contaminated by effects of $\ell' + 1$ treatment periods

$\Rightarrow \Delta_g(\ell) = \Delta_{g'}(\ell) \forall g, g'$ **not sufficient** for $\hat{\beta}_\ell$ to be unbiased. We also need

$$\Delta_g(\ell') = \Delta_{g\hat{a}}(\ell') \quad \forall \ell' \neq \ell$$

Origin: Again, Forbidden Comparisons

- As in the static case, $\hat{\beta}_\ell$ can still be written as a linear combination of DID
- But in some of these DIDs:
 - The control group has been treated for $\ell' \neq \ell$ period at baseline/endline
 - If $\ell > 0$, the treated group has been treated for $\ell' > 0$ period at the baseline
 - The treated group has been treated for $\ell' < \ell$ period at the endline

⇒ Contamination bias

A Simple Example

- Consider a design with $G = 2$, $T = 3$, $K = 0$
 - Group 1 treated at $t = 2$
 - Group 2 treated at $t = 3$
- Then, some algebra shows that:

$$\hat{\beta}_1 = \underbrace{Y_{1,3} - Y_{1,2} - (Y_{2,3} - Y_{2,2})}_{DID_1} + 2 \underbrace{[Y_{1,2} - Y_{1,1} - (Y_{2,2} - Y_{2,1})]}_{DID_2}$$

A Simple Example (contâd)

- In DID_1 , control group $g = 2$ treated for 1 period at $t = 3$ and $g = 1$ also treated for 1 period at $t = 2$. Then,

$$E[DID_1] = E[\Delta_1(1) - (\Delta_1(0) + \Delta_2(0))].$$

- DID_2 identifies the effect of having been treated for 1 rather than 2 periods:

$$E[DID_2] = E[\Delta_1(0)]$$

⇒ We obtain $E[\hat{\beta}_1] = E[\Delta_1(1) + (\Delta_1(0) - \Delta_2(0))]$

Software Implementations

- eventstudyweights Stata package computes weights in decomposition:

```
eventstudyweights {rel_time_list}, absorb(i.groupid i.timeid)  
cohort(first_treatment) rel_time(ry),
```

with $\text{rel_time_list} = \text{list of the } 1\{F_g = t - l\}$, $\text{first_treatment} = F_g$
(missing if never treated), and $\text{ry} = \text{timeid} - \text{first_treatment}$

- Can easily include covariates

Heterogeneity-robust DID estimators

Robust DIDs (Static Case)

- Avoid making the **forbidden comparisons** leveraged by TWFE:
 - ⇒ Never compare switcher to switcher: only compare switcher to stayer
 - ⇒ Never compare a switcher to a stayer with a different baseline treatment
- Depends on whether groups' outcome at t only depends on treatment at t
- If yes, we can **consider each pair of consecutive time periods independently**
 - ⇒ $t - 1$ to t switchers: groups whose treatment changes from $t - 1$ to t
 - ⇒ $t - 1$ to t stayers: groups whose treatment does not change from $t - 1$ to t , with same $t - 1$ treatment as switchers

Robust DIDs (Dynamic Case)

- in more general case, we need to **control for groups' full treatment history**
 - ⇒ $t - 1$ to t first-time switchers: treat changes for the first time from $t - 1$ to t
 - ⇒ 1 to $t + \ell$ stayers: treat does not change from 1 to $t + \ell$, with same $t - 1$ treatment as switchers
- Allowing for **dynamic effects** may lead to less precise and interpretable effects

Parameters of interest

- Suppose first that D is binary
- Let us define

$$\mathcal{S} = \{(g, t) : t \geq 2, D_{g,t} \neq D_{g,t-1}, \exists g' : D_{g',t} = D_{g',t-1} = D_{g,t-1}\}$$

- $N_S = \text{card}(\mathcal{S})$

⇒ Then, ATE for **matchable switchers** is

$$\delta^S = E \left[\frac{1}{N_S} \sum_{(g,t) \in \mathcal{S}} Y_{g,t}(1) - Y_{g,t}(0) \right]$$

Assumptions for identifying δ^S

- δ^S can be unbiasedly estimated under the following // trends conditions

$$\Rightarrow E[Y_{g,t}(0) - Y_{g,t-1}(0) | \mathbf{D}_g] = E[Y_{g,t}(0) - Y_{g,t-1}(0)] = \gamma_{0,t}$$

$$\Rightarrow E[Y_{g,t}(1) - Y_{g,t-1}(1) | \mathbf{D}_g] = E[Y_{g,t}(1) - Y_{g,t-1}(1)] = \gamma_{1,t}$$

- Usual // trends on $Y_{g,t}(0)$ sufficient if we focus on switchers in

$$\mathcal{S}_+ = \{(g, t) : t \geq 2, D_{g,t} = 1 > D_{g,t-1} = 0, \exists g' : D_{g',t} = D_{g',t-1} = 0\}$$

- Weaker exogeneity assumption sufficient to consistently estimate δ^S :

$$E[Y_{g,t}(0) - Y_{g,t-1}(0) | D_{g,1}, \dots, D_{g,t}] = E[Y_{g,t}(0) - Y_{g,t-1}(0)]$$

- Allows for possibility that $Y_{g,t}(0) - Y_{g,t-1}(0)$ affects $D_{g,t+1}$ etc.

Weighted averages of DIDs identify δ^S

- For all $t \in \{1, \dots, T\}$ and $d = 0, 1$, let
 - $N_{+,t} = \text{card} \{g : D_{g,t} > D_{g,t-1}\}$ and $N_{-,t} = \text{card} \{g : D_{g,t} < D_{g,t-1}\}$
 - $N_{=d,t} = \text{card} \{g : D_{g,t} = D_{g,t-1} = d\}$
 - $DID_{+,t} = \sum_{g:D_{g,t} > D_{g,t-1}} \frac{1}{N_{+,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t} = D_{g,t-1} = 0} \frac{1}{N_{=0,t}} (Y_{g,t} - Y_{g,t-1})$
 - $DID_{-,t} = \sum_{g:D_{g,t} = D_{g,t-1} = 1} \frac{1}{N_{=1,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t} < D_{g,t-1}} \frac{1}{N_{-,t}} (Y_{g,t} - Y_{g,t-1})$

- Then

$$E[DIDM] = E \left[\sum_{t=2}^T \frac{N_{+,t}}{N_S} DID_{+,t} + \frac{N_{-,t}}{N_S} DID_{-,t} \right] = \delta^S$$

Intuition for *DIDM*

- $DID_{+,t}$: ΔY between those treated between $t - 1$ and t , and untreated
- ⇒ Under // trends on $Y(0)$, it identifies TE in groups switching into treatment
- ⇒ Under // trends on $Y(1)$, $DID_{-,t}$ identifies TE for switchers out of treat
- Finally, *DIDM* is a weighted average of those DID estimands

Placebo Estimators

- ⇒ Focus on groups that are stayers one period before switchers switch
- Compare switchers' and stayers' ΔY , one period before switchers switch
 - Also compare switchers and stayers 2, 3 periods etc. before switchers switch

Controlling for Time-varying Covariates

- Rationale: // trends only hold if we account for **change in covariates**

$$E(Y_{g,t}(d) - Y_{g,t-1}(d) | \mathbf{D}_g, \mathbf{X}_g) = \gamma_{d,t} + (X_{g,t} - X_{g,t-1})' \lambda_d \quad \forall d \in \mathcal{D}$$

- Special case is group-specific linear trends $X_{g,t} = (1\{g=2\} \times t, \dots, 1\{g=G\} \times t) \hat{a}$
- Let $\epsilon_{g,t}(d)$ residual of the reg. of $Y_{g,t} - Y_{g,t-1}$ on period FEs and $X_{g,t} - X_{g,t-1}$ for (g,t) s.t. $D_{g,t} = D_{g,t-1} = d \in \mathcal{D}$
- Define ***DIDM^X*** as *DIDM*, but using $\epsilon_{g,t}(D_{g,t-1})$ instead of $Y_{g,t} - Y_{g,t-1}$
- Separate reg. for each $d \in \mathcal{D}$, estimated in **sample of *d*-stayers**

Software Implementation

- R and Stata command: `did_multiplegt`
 - Options to relax the standard `//` trends
 - Control for time-varying, time-invariant covariates, or linear time trends
 - Flexibly specifies the number of placebos to be estimated

Example (continued): Gentzkov et al. (AER, 2011)

Table: Estimates of the effect of one additional newspaper on turnout

	Estimate	Standard error	N
$\hat{\beta}_{fd}$	0.0026	0.0009	15,627
$\hat{\beta}_{fe}$	-0.0011	0.0011	16,872
<i>DIDM</i>	0.0043	0.0014	16,872
<i>DIDM</i> Placebo	-0.0009	0.0016	13,221

⇒ *DIDM* is 66% larger (t-stat=1.77) than $\hat{\beta}_{fd}$ and opposite sign to $\hat{\beta}_{fe}$

Allowing for dynamic effects: potential outcome notation

- We maintain previous assumption of **no anticipation effects**
- ⇒ Potential outcomes $Y_{g,t}(d_1, \dots, d_t)$ do not depend on future treatments
- $\mathbf{0}_k$: vector of k zeros, $\mathbf{1}_k$: vector of k ones
 - $Y_{g,t}(\mathbf{0}_k, \mathbf{1}_{t-k})$: g at t if untreated from 1 to k and treated from $k+1$ to t

Parameters of interest

- **Binary staggered design**: $D_{g,t} = 1\{t \geq F_g\}$

⇒ **ATE of cohort getting treated** from c to $c + l$, i.e. for $l + 1$ periods

$$\delta(c, \ell) = E \left[\frac{1}{N_{=}(c)} \sum_{g: F_g = c} (Y_{g, c+\ell}(\mathbf{0}_{c-1}, \mathbf{1}_{\ell+1}) - Y_{g, c+\ell}(\mathbf{0}_{c+\ell})) \middle| \mathbf{D} \right]$$

- Where $N_{=}(c) = \text{card}\{g : F_g = c\}$
- **ATE across cohorts** $\ell + 1$ periods after they started receiving treat

$$\delta(\ell) = E \left[\frac{1}{\sum_{c=2}^{T-\ell} N_{=}(c)} \sum_{c=2}^{T-\ell} N_{=}(c) \delta(c, \ell) \middle| \mathbf{D} \right]$$

DID estimators using never-treated groups

- To estimate $\delta(c, \ell)$

$$\hat{\delta}_1(c, \ell) = \frac{1}{N_{=}(c)} \sum_{g:F_g=c} (Y_{g,c+\ell} - Y_{g,c-1}) - \frac{1}{N_{=}(T+1)} \sum_{g:F_g=T+1} (Y_{g,c+\ell} - Y_{g,c-1})$$

- Recall that $\{g : F_g = T + 1\} =$ **never-treated groups**

$\Rightarrow \hat{\delta}_1(c, \ell)$: ΔY between $c - 1$ and $c + \ell$ in cohort c and in never-treated groups

- Then let

$$\hat{\delta}_1(\ell) = \frac{1}{\sum_{c=2}^{T-\ell} N_{=}(c)} \sum_{c=2}^{T-\ell} N_{=}(c) \hat{\delta}_1(c, \ell)$$

DID estimators using other control groups

- If no never-treated groups, **use instead the last treated/not-yet-treated groups**

$$\hat{\delta}_2(c, \ell) = \frac{1}{N_=(c)} \sum_{g: F_g = c} (Y_{g, c+\ell} - Y_{g, c-1}) - \frac{1}{N_>(c+\ell)} \sum_{g: F_g > c+\ell} (Y_{g, c+\ell} - Y_{g, c-1})$$

- $N_>(c+\ell) = \text{card}\{g : F_g > c+\ell\}$. Define $\hat{\delta}_2(\ell)$ accordingly
- There are **more not-yet-treated groups** than never-treated groups

$\Rightarrow \hat{\delta}_2(c, \ell)$ possibly more precise than $\hat{\delta}_1(c, \ell)$

- Also, never-treated groups may be very different from other groups

$\Rightarrow //$ **trends may not hold when only never treated** used as controls

Software Implementation

- The `csdid` (Stata) and `did` (R) commands. Syntax:

```
csdid outcome, time(timeid) gvar(cohort)
```

where $\text{cohort} = F_g$ ($=0$ for never treated)

- The `eventstudyinteract` Stata command. Syntax:

```
eventstudyinteract outcome {rel_time_list}, absorb(i.groupid  
i.timeid) cohort(first_treatment) control_cohort(controlgroup)
```

A Quick Recap on Job Search Models

The Optimal Stopping Model

- Risk neutral individual in discrete time with preferences in $t = 0$ given by

$$\sum_{t=0}^{\infty} \beta^t c_t, \quad \beta \in (0, 1)$$

- Start as unemployed, with consumption equal to b
 - **Jobs sampled sequentially.** Each job is for life and identical except for wage
 - Wages are drawn from an exogenous stationary distribution $F(w)$
- ⇒ Given draw $w_t \in W$ **agent decides whether to take it or continue searching**

Dynamic Programming Formulation

- Value function for the agent when he has sampled a job of $w \in W$ is

$$V(w) = \max \left\{ \frac{w}{1-\beta}, \beta V + b \right\}$$

where V is the **continuation value of not accepting** a job:

$$V = \int_{\omega \in \Omega} V(\omega) dF(\omega)$$

⇒ Combine these two equations and get:

$$V(w) = \max \left\{ \frac{w}{1-\beta}, b + \beta \int_{\omega \in \Omega} V(\omega) dF(\omega) \right\}$$

Reservation Wage

- $V(w)$ is non-decreasing, **decision rule has a reservation value** property
- **Reservation wage** is given by

$$\frac{w^*}{1-\beta} = b + \beta \int_{\omega \in \Omega} V(\omega) dF(\omega)$$

⇒ Decision rule: $\forall w < w^*, V(w) = \frac{w^*}{1-\beta}$ and $\forall w \geq w^*, V(w) = \frac{w}{1-\beta}$

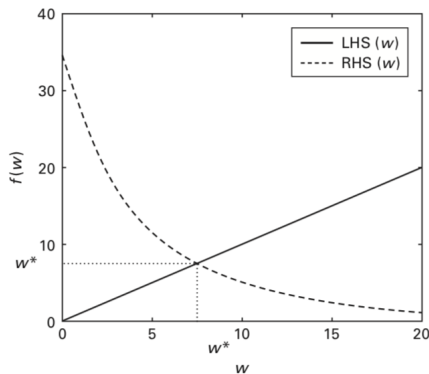
- Therefore, reservation wage can be written as

$$\frac{w^*}{1-\beta} = b + \beta \left[\frac{w^* F(w^*)}{1-\beta} + \int_{w \geq w^*} \frac{w}{1-\beta} dF(w) \right]$$

Reservation Wage

- Since $\frac{w^*}{1-\beta} = \int_{w < w^*} \frac{w^*}{1-\beta} dF(w) + \int_{w \geq w^*} \frac{w^*}{1-\beta} dF(w)$

$$w^* = b + \frac{\beta}{1-\beta} \left[\int_{w \geq w^*} (w - w^*) dF(w) \right]$$



Taking the Model to the Data

- For a random sample of N workers we observe $\{\tilde{t}_u(i), w(i)\}_{i=1}^N$
- Job offers/termination arrive at random times with density between offers

$$q_u(t_u) = \lambda \exp(-\lambda t_u), \lambda > 0$$

$$q_e(t_e) = \eta \exp(-\eta t_e), \eta > 0$$

- Reservation wage in **continuous time**

$$w^* = b + \frac{\lambda}{\rho + \eta} \int_{w^*} (w - w^*) dF(w)$$

⇒ It is easy to show that $\partial w^* / \partial \eta < 0$, $\partial w^* / \partial \rho < 0$, $\partial w^* / \partial \lambda > 0$

Simulating Labor Market Data

Hypothetical early labor market career

Event number	State	Time of event	Duration draw	Match value
1	<i>U</i>	0.891	0.891	6.243
2	<i>U</i>	3.168	2.277	4.329
3	<i>U</i>	15.554	12.386	3.871
4	<i>U</i>	15.558	0.004	10.918
5	<i>E</i>	38.921	23.363	—
6	<i>U</i>	44.236	5.315	7.891
7	<i>U</i>	56.793	12.557	12.119
8	<i>E</i>	157.421	100.628	—
9	<i>U</i>	164.772	7.351	10.145
10	<i>E</i>	322.510	157.738	—
⋮	⋮	⋮		⋮

Steady-state Proportions

- The probability that an individual is unemployed is

$$\begin{aligned} p(u) &= \frac{\mathbb{E}(t_u)}{\mathbb{E}(t_e) + \mathbb{E}(t_u)} = \frac{[\lambda \tilde{F}(w^*)]^{-1}}{\eta^{-1} + [\lambda \tilde{F}(w^*)]^{-1}} \\ &= \frac{\eta}{\eta + \lambda \tilde{F}(w^*)} \end{aligned}$$

- And conversely

$$p(e) = 1 - p(u) = \frac{\lambda \tilde{F}(w^*)}{\eta + \lambda \tilde{F}(w^*)}$$

Offered and Accepted Wages

- Optimal decision rule $\Rightarrow$ **truncation in the accepted wage** distribution

$$g(w) = \frac{f(w)}{\tilde{F}(w^*)}, w \geq w^*$$

- This density is well defined: integrates to 1 and non-negative for all $w \geq w^*$

Likelihood Contributions

- The likelihood of an ongoing unemployment spell is

$$L(t_u, u) = f_u(t_u)p(u) = \lambda\tilde{F}(w^\star) \exp[-\lambda\tilde{F}(w^\star)t_u] \times \frac{\eta}{\eta + \lambda\tilde{F}(w^\star)}$$

- The likelihood of employed and earning a wage w is

$$L(w, e) = \frac{f(w)}{\tilde{F}(w^\star)} \times \frac{\lambda\tilde{F}(w^\star)}{\eta + \lambda\tilde{F}(w^\star)} = \frac{\lambda f(w)}{\eta + \lambda\tilde{F}(w^\star)}$$

Likelihood Function

- The likelihood function for a random sample of N individuals is then

$$L(w_1, \dots, w_{N_e}, t_1, \dots, t_{N_u}) = \prod_{i \in e} \left[\frac{\lambda f(w(i))}{\eta + \lambda \tilde{F}(w^*)} \right] \times \prod_{i \in u} \left[\frac{\eta \lambda \tilde{F}(w^*) \exp[-\lambda \tilde{F}(w^*) t_u(i)]}{\eta + \lambda \tilde{F}(w^*)} \right]$$

- And the associated log-likelihood is

$$\begin{aligned} \ln L = & -N \ln[\eta + \lambda \tilde{F}(w^*)] + N \ln \lambda + \sum_{i \in e} \ln[f(w(i))] + \\ & + N_u \ln[\tilde{F}(w^*)] + N_u \ln(\eta) - \lambda \tilde{F}(w^*) \sum_{i \in u} t_u(i) \end{aligned}$$

Identification

- The primitive parameters that explicitly enter in $\ln L$ are λ, η and F
- Parameters b and ρ only enter through w^*
- Equilibrium object w^* is part of the support of F

⇒ This feature generates a non-standard likelihood function

Flinn and Heckman (1982)

- 1 Estimate the reservation wage as the **minimum accepted wage**

$$\hat{w}^* = \min(w_1, \dots, w_{N_e})$$

⇒ Order statistics are super-consistent (i.e. converge at rate N)

- 2 Maximize log likelihood with respect to λ, η and μ **conditional on $\hat{w}^*$**

⇒ $F(w)$ needs to be recoverable: $F(w|w \geq \hat{w}^*) = \frac{F(w) - F(w^*)}{F(w^*)}, \forall w \geq \hat{w}^*$

- 3 Plug estimated parameters into **equation for w^* and solve** for either b or ρ

⇒ Usually fix ρ and recover the value of b

Informality, job search, and schooling investments

Bobba, Flabbi and Levy (IER, 2022)

- An **equilibrium search model** where:
 - Search frictions generate mobility between formal and informal jobs
 - Match productivity and bargaining generate overlapping wage distributions
 - ⇒ Both ingredients generates a mix of formal and informal jobs in equilibrium
- One long-term “cost of informality”: **Under-investment in education**
 - Same features that create informality may also distort returns to schooling
 - ⇒ Trade-off between welfare in the labor market and pre-market HK

Context: Labor Markets in Latin America

- ① More than half of the labor force is in the **informal sector**
 - Workers not contributing to and not covered by the social security system
 - ⇒ Informal employees and (most of the) self-employed
- ② Neither a **segmented or a competitive** labor market
 - Individuals transit back and forth between formal and informal jobs
 - Wage/productivity distributions overlap
 - Mix of formality status within the same firm
- ③ Informal workers gained access to **non-contributory social programs**

The Model Environment

- Timing
 - 1 Schooling decision
 - 2 Searching status decision
 - 3 Labor market dynamics
- Labor Market States
 - 1 Unemployed
 - 2 Self-employed
 - 3 Informal Employee
 - 4 Formal Employee

Schooling Decision

- Irrevocable decision about schooling level $h \in \{0, 1\}$
- Individual-specific heterogeneity
 - costs $\kappa \sim T(\kappa)$
 - opportunity cost - PDV of participating in LMK as $h = 0$

⇒ Only agents with $\kappa < \kappa^*(y)$ will acquire $h = 1$

- All labor market parameters are allowed to be schooling-specific

Searching-status Decision

- Irrevocable decision $s \in \{0, 1\}$:
 - Self-employed ($s = 1$)
 - Unemployed ($s = 0$)
 - Search for a job in both states but receive offers at different rates: $\gamma_h < \lambda_h$
 - Self-employment income $y \sim R(y|h)$
- ⇒ Only agents with $y \geq y^*(h)$ search while also working as self-employed

Labor Market Dynamics

State	PDV	Shock	Flow Utility
Workers:			
Unemployed	$U(h)$	λ_h	$\xi_h + \beta_{0,h}B_0$
Self-Employed	$S(y, h)$	γ_h	$y + \beta_{0,h}B_0$
Informal Employee	$E_0[w, y, h]$	η_h, χ_h	$w_0(x; y, h) + \beta_{0,h}B_0$
Formal Employee	$E_1[w, y, h]$	η_h, χ_h	$w_1(x; y, h) + \beta_{1,h}B_1[w_1(x; y, h)]$
Firms:			
Vacancy	$V[h]$	ζ_h	ν_h
Filled Informal Job	$F_0[x, y, h]$	η_h, χ_h	$x - w_0(x; y, h)$
Filled Formal Job	$F_1[x, y, h]$	η_h, χ_h	$x - (1 + t)w_1(x; y, h)$

⇒ Match-specific productivity: $x \sim G(x|h)$

⇒ One-shot penalty for firms hiring illegally: $c_h w_0(x; y, h)$

⇒ Matching function determines $\{\lambda_h, \gamma_h, \zeta_h\}$: $m_h = (u_h + \psi_h s_h)^{\iota_h} (\nu_h)^{1-\iota_h}$

Labor Market Institutions and Wage Determination

- Non-wage workers' flow value:

- formal employee $= \beta_{1,h} B_1 [w_1(x; y, h)] = \beta_{1,h} [\tau t w_1(x; y, h) + b_1]$
- informal employee $= \beta_{0,h} B_0$

$\Rightarrow b_1$ generates spillovers within and between schooling levels

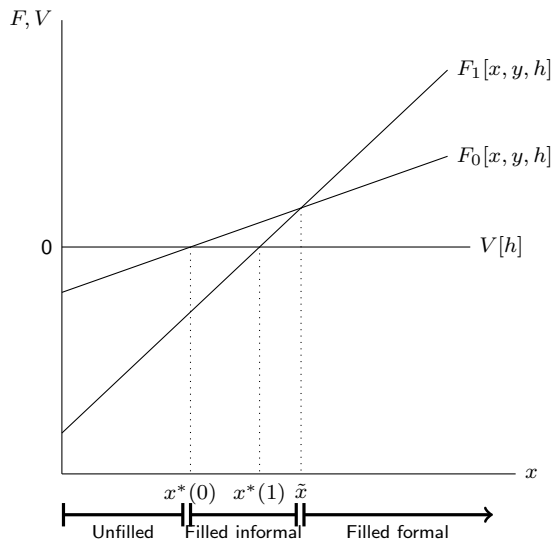
- Nash-bargaining wage schedules (under free-entry of firms) are:

$$w_0(x; y, h) = \frac{\alpha_h}{1 + \chi_h c_h} x + (1 - \alpha_h) [\rho Q(y, h) - \beta_{0,h} B_0]$$

$$w_1(x; y, h) = \frac{\alpha_h}{1 + t} x + \frac{(1 - \alpha_h)}{1 + \beta_{1,h} \tau t} [\rho Q(y, h) - \beta_{1,h} b_1]$$

$\Rightarrow Q(y, h) \equiv \max\{S(y, h), U(h)\}$

Equilibrium Representation



Empirical Implications

⇒ Main stylized facts of informal labor markets are replicated in equilibrium:

- 1 A mixture of formal and informal jobs is realized
- 2 Formal employees have on average higher wages than informal employees
- 3 But their accepted wage distributions overlap
- 4 Informal employees and self-employed have different labor market dynamics
- 5 Firms hire formal or informal workers at different points in time
- 6 Workers transit over time between different formality status

Data Sources

1 Mexico's Labor Force Survey (ENOE) in 2005

- Nonagricultural, full-time, male, private-sector workers
- ⇒ Secondary-school between the ages of 25 and 55 who reside in urban areas
- $w \equiv$ Hourly wages as employee, main job after labor contributions
- $y \equiv$ Hourly labor income as self-employed, without paid employees
- $f = 1$ if employee is contributing to the social-security fund; $= 0$ otherwise
- $h = 1$ if Upper secondary completed $= 0$ if Lower secondary completed

2 Aggregate labor shares for Mexico in 2005

- Total compensations per employee as percentage of GDP

3 Vacancy rates for 2005

- Good coverage of vacancy posting in urban areas
- ⇒ Detailed information on the schooling level required for the job

Identification: Search-Matching-Bargaining Parameters

- $G(x|h)$: Has to be “recoverable”
 - ⇒ We assume lognormal with parameters $\{\mu_{x,h}, \sigma_{x,h}\}$
- $\lambda_h, \gamma_h, \eta_h$: stationarity + optimal decision rules identify mobility rates from
 - ⇒ Transitions
 - ⇒ Steady state distributions over labor market states
- ρ, ξ_h : Use $Q(y, h)$ to obtain their joint identification
- Nash Bargaining coefficient: $\alpha_1 = \alpha_0 = \alpha$
 - ⇒ Use labor shares ($w_f(x; y, h)/x$)

Identification: Matching Function + Demand Side

- $\{\psi_h, \iota_h\}$: use vacancy rate and define mkt tightness $\omega_h \equiv \frac{v_h}{u_h + \psi_h s_h}$, so that

$$\begin{aligned}\psi_h &= \frac{\gamma_h}{\lambda_h} \\ \iota_h &= \frac{\ln \omega_h - \ln \lambda_h}{\ln \omega_h}\end{aligned}$$

- Then, back out the demand side parameters
 - $\zeta_h = \omega_h^{-\iota_h}$
 - ν_h : use firm's value function and impose free entry

Identification: Informality Parameters (β_1 and c_h)

- Different transition rates out of formal jobs and informal jobs identify χ_h
- Overlap between formal and informal accepted wage distributions

$$w_0(\tilde{x}(y, h); y, h) - w_1(\tilde{x}(y, h); y, h) > 0$$

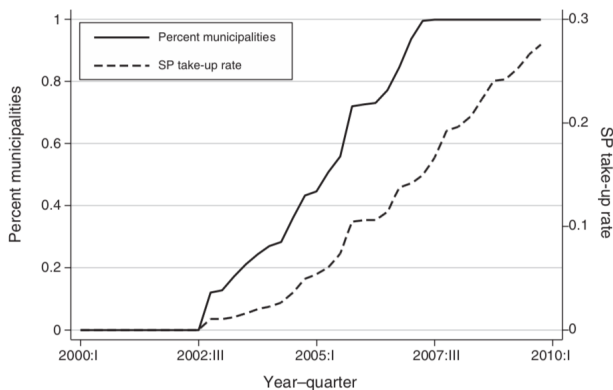
⇒ Given x , formal employees receive higher non-wage benefits

⇒ β_1 and c_h alter **location and overlap** of accepted wage distributions

- Variation in y is useful variation to separately identify the parameters

Identification: Informality Parameters (β_0)

- The identification of β_0 requires the use of additional information
- We exploit **staggered entry of the Seguro Popular (SP) program in 2005**



⇒ In terms of our model, $SP \approx \uparrow$ in B_0 by 25%

Identification: Informality Parameters (β_0 , cont'd)

- Variation in B_0 identify β_0 if uncorrelated with changes in model primitives

⇒ Labor market outcomes pre-policy (2002) seem balanced

	Hourly Wages (log)			Labor Market Proportions			
	Formal	Informal	Self	Formal	Informal	Self	Unempl
SP in 2005 (1=yes)	-0.041 (0.036)	0.048 (0.055)	-0.035 (0.062)	-0.034 (0.026)	0.035 (0.019)	-0.004 (0.014)	0.003 (0.006)
Complete Sec. (1=yes)	0.218 (0.017)	0.288 (0.032)	0.092 (0.033)	0.061 (0.011)	-0.036 (0.008)	-0.029 (0.008)	0.003 (0.003)
Number of Obs.	7865	5474	2777	16458	16458	16458	16458

Identification: Self-employment and Schooling Parameters

- $R(y|h)$: Identified by observed self-employment earnings, once we assume a recoverable primitive distribution
 - ⇒ We assume lognormal with parameters $\{\mu_{y,h}, \sigma_{y,h}\}$
- $T(\kappa)$: The threshold crossing decision rule allows for the identification of one parameter from the proportions of individuals in the two schooling levels

$$\frac{1}{n} \sum_{i=1}^n h_i = \int_y T(\kappa^*(y)) dR(y|0)$$

- ⇒ We assume a negative exponential with parameters δ

Identification: Unobserved Ability Types

- Type is known to the individual but unobserved in the data
- We denote each type with k and its proportion in the population with π_k

$$x|k = a_k^G x$$

$$y|k = a_k^R y$$

$$\kappa|k = a_k^T \kappa$$

⇒ Duration dependence in unemployment identifies these parameters

- Hazard rates at three and six months for both schooling levels
- Assume $K = 2$
 - type $k = 1$ normalized to $a_1^T = a_1^R = a_1^G = 1$
 - type $k = 2$ exhibiting $a_2^T < 1; a_2^R > 1; a_2^G > 1$

Estimation in Two Steps

- ① For $s \in \{0, 1\}$ and $SP \in \{0, 1\}$, we match the following moments
 - Proportions of individuals in each labor market state
 - Accepted wage distributions of formal and informal employees
 - ⇒ Mean and SD: overall and by quintiles
 - ⇒ Overlap: % of formal empl. for each quintile of the informal wage distribution
 - Accepted earnings distributions of self-employed
 - ⇒ Mean and SD
 - Transitions between LMK states (yearly)
 - Hazard rates out of unemployment (at 3 and 6 months)
 - Labor Shares
- ② Estimate demand-side parameters using vacancy rates

Parameter Estimates (selected coeffs)

	Low Schooling $h = 0$		High Schooling: $h = 1$	
	Coeff.	Std. Error	Coeff.	Std. Error
Search, Matching, and Bargaining				
λ_h	0.4679	0.0035	0.5167	0.0098
γ_h	0.0349	0.0042	0.0306	0.0014
η_h	0.0326	0.0007	0.0190	0.0052
$\mu_{x,h}$	2.7616	0.0367	2.6749	0.0382
$\sigma_{x,h}$	0.6243	0.0132	0.7970	0.0038
$\mu_{y,h}$	1.6718	0.0188	1.9497	0.0763
$\sigma_{y,h}$	0.7754	0.0028	0.8027	0.0258
ξ_h	-103.46	1.6661	-158.05	4.6038
α	0.5630	0.0169	0.5630	0.0169
Preferences and Informality				
$\beta_{1,h}$	0.7949	0.0044	0.6091	0.0043
$\beta_{0,h}$	0.9862	0.0038	0.9807	0.0015
χ_h	0.0079	0.0004	0.0113	0.0008
c_h	12.882	0.7045	16.574	1.3932
Matching Function and Demand Side				
ψ_h	0.0745	0.0088	0.0592	0.0034
ι_h	0.7321	0.0253	0.7281	0.0184
ζ_h	7.9718	1.6278	5.8569	0.8742
ν_h	-496.01	288.80	-773.80	111.34

Returns to Schooling

	Ability:	Low	High
		$k = 1$	$k = 2$
<u>PDV of Labor Market Search:</u>			
$\int_y Q(y, h) dR(y h)$		0.309	0.278
<u>Average Accepted Wages:</u>			
F: $E_h [w_1 \mid \tilde{x}(y, h) \leq x]$		0.479	0.435
I: $E_h [w_0 \mid x_0^*(y, h) \leq x < \tilde{x}(y, h)]$		0.281	0.296
<u>Average Offered Wages:</u>			
F: $E_h [w_1 \mid y < y^*(h)]$		0.213	0.210
F: $E_h [w_1 \mid y \geq y^*(h)]$		0.213	0.204
I: $E_h [w_0 \mid y < y^*(h)]$		0.133	0.134
I: $E_h [w_0 \mid y \geq y^*(h)]$		0.142	0.136

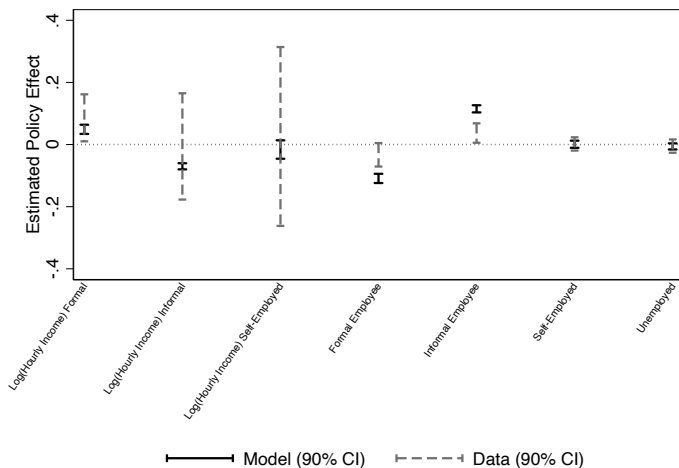
Out-of-Sample Model Validation

- Estimate the effect of $\uparrow B_0$ using SP roll-out one year later (2006)

$$y_{i,q} = \theta d_{m(i),q} + \vartheta h_i + \varrho_{m(i)} + \varphi_q + \epsilon_{i,q}$$

- Predict change in LMK outcomes with B_0^{2006} using estimated model
- Estimate TWFE-DID specifications on both actual and simulated data

Out-of-Sample Model Validation

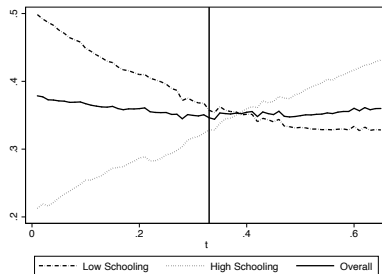


Counterfactual 1: The Equilibrium Effects of Informality

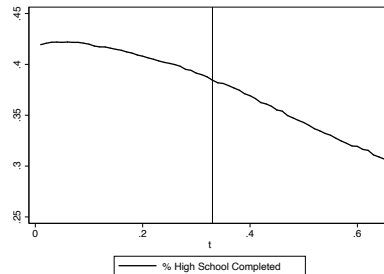
Model:	Firms can only offer a formal contract			
Specifications:	Baseline Model	Exogenous Schooling	Exogenous Contact Rates	Hosios-like Condition ($\alpha = \iota$)
<u>Flow Welfare:</u>				
Total	-0.0596	-0.0750	-0.0020	0.0478
Workers	-0.0460	-0.0599	0.0166	0.0570
Firms	-0.2821	-0.3219	-0.3055	-0.1589
<u>Labor Market Proportions:</u>				
Unemployed	0.0213	0.0636	0.0019	-0.0459
Self-employed	0.3353	0.3526	0.3625	0.2329
Formal Employees	0.0275	-0.0146	-0.0376	0.0076
<u>Schooling Outcomes:</u>				
% HS Completed	0.1029	—	0.0781	0.1501
% High Ability in HS	0.0538	—	0.0569	0.0628

NOTE: Relative changes wrt the benchmark model. Hosios increases α from 0.56 to 0.73.

Counterfactual 2: Changes in Payroll Tax Rate (t)



(a) Informality



(b) Schooling

- Composition effects over schooling/ability explain no impact on informality
- Balanced-budget policy with $\tau = 0 \rightarrow 10\%$ increase in high-school completion

Main Takeaways from the Estimated Model

- ① Returns to schooling are substantial
- ② Informality is welfare improving but:
 - Significantly more so for firms than workers
 - Reduces human capital accumulation (hold-up problem)
- ③ Payroll tax rate has a non-intuitive impact on equilibrium outcomes
 - Informality rate not a good indicator for policy
 - Redistributive forces within the formal system are key

DID $\Leftrightarrow$ Economic Model

- Relevant institutional features are included in the model in a tractable way
- These parameters are hard to separately identify using labor market data
- The staggered roll-out of the policy provides additional variation to:
 - $\Rightarrow$ Identify the (average) valuation of non-contributory benefits
 - $\Rightarrow$ Validate the model on a different time period by simulating one-step ahead